AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

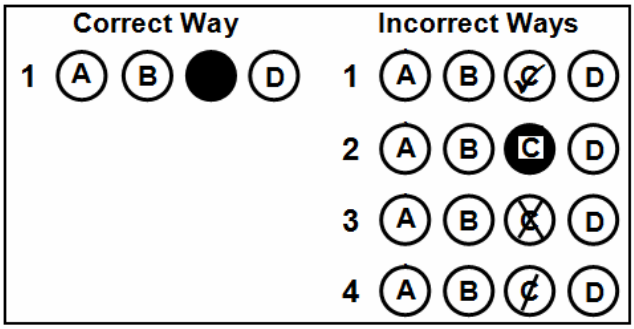
MAY 2012

General Mathematics Paper I

Time allowed: 40 minutes Marks 30

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a simple calculator if you wish.

1. If $x = \frac{1}{2}$, then the value of $(2x + a)$ is equal to

   A. a
   B. $\frac{a}{2}$
   C. $a + \frac{1}{2}$
   D. $a + 1$

2. If $(a+b)^2 = 4$ and $(a-b)^2 = 2$, then the value of ab is equal to

   A. $\frac{1}{2}$
   B. 1
   C. 2
   D. -2

3. If $(ax - 2y - bz)^2 = x^2 + 4y^2 + 9z^2 - 4xy + 12yz - 6xz$, then the values of a and b are equal to

   A. -1 and 3
   B. 1 and 3
   C. 1 and -3
   D. -1 and -3

4. Which of the following products is equal to $(x - y)$?

   A. $\left(\sqrt{x} - \sqrt{y}\right)\left(\sqrt{x} - \sqrt{y}\right)$
   B. $\left(\sqrt{y} - \sqrt{x}\right)\left(\sqrt{y} + \sqrt{x}\right)$
   C. $\left(\sqrt{y} - \sqrt{x}\right)\left(\sqrt{y} - \sqrt{x}\right)$
   D. $\left(\sqrt{x} - \sqrt{y}\right)\left(\sqrt{x} + \sqrt{y}\right)$

5. The factors of $\left(\frac{1}{x^3} - 1\right)$ are

   A. $\left(\frac{1}{x} - 1\right)$ and $\left(\frac{1}{x^2} + \frac{1}{x} + 1\right)$
   B. $\left(\frac{1}{x} - 1\right)$ and $\left(\frac{1}{x^2} - \frac{1}{x} + 1\right)$
   C. $\left(\frac{1}{x} + 1\right)$ and $\left(\frac{1}{x^2} - \frac{1}{x} + 1\right)$
   D. $\left(\frac{1}{x} + 1\right)$ and $\left(\frac{1}{x^2} + \frac{1}{x} + 1\right)$

6. If $(x + 1)$ is a factor of the expression $\left(k - x^2 + x\right)$, then the value of k is equal to

A. -2
B. 0
C. 2
D. 3

7. The Highest Common Factor (H.C.F) of $15x^2$ and $45xy$ is

A. $15x$
B. $15x^2$
C. $45xy$
D. $45xy^2$

8. If $P(x) = x(3 - x)^2$, the Least Common Factor (L.C.M) $= x^2(3 - x)$ and the Highest Common Factor (H.C.F) $= (3 - x)$, then the other polynomial will be

A. 1
B. x
C. $x(3 - x)$
D. $\dfrac{(3 - x)}{x}$

9. $\sqrt{\dfrac{1}{a^2} + \dfrac{1}{b^2} - \dfrac{2}{ab}}$ is equal to

A. $\dfrac{1}{a} - \dfrac{1}{b}$
B. $\dfrac{1}{a} + \dfrac{1}{b}$
C. $\dfrac{1}{a} + \dfrac{1}{b} - \dfrac{1}{ab}$
D. $\dfrac{1}{a} + \dfrac{1}{b} - \sqrt{\dfrac{2}{ab}}$

10. If $\sqrt{\dfrac{5 - 2x}{2x}} = 0$, then the value of x is equal to

A. $\dfrac{2}{5}$
B. $\dfrac{4}{5}$
C. $\dfrac{5}{2}$
D. $\dfrac{5}{4}$

11. Which of the following equations has the solution set $\{-6, 2\}$?

A. $|x + 2| = 4$
B. $|x - 2| = 4$
C. $|x + 2| = -4$
D. $|x - 2| = -4$

12. If $2x + 2 = 11$, then the value of $\sqrt{2x + 7}$ is equal to

A. 3
B. 4
C. 5
D. 6

13. If $px^2 - 5x + 6 = \left(x - \frac{5}{2}\right)^2 - \frac{1}{4}$, then the value of p is equal to

A. 4
B. 2
C. 0
D. 1

14. Which of the following equations is NOT a quadratic equation?

A. $x - \frac{3}{x} = 3$
B. $3x - \frac{5}{x} = x^2$
C. $x + \frac{1}{x} = 3$
D. $x^2 - 3 = 4x^2 - 4x$

15. The additive inverse of the matrix $\begin{bmatrix} -1 & 3 \end{bmatrix}$ is

A. $\begin{bmatrix} -1 & -3 \end{bmatrix}$
B. $\begin{bmatrix} 1 & -3 \end{bmatrix}$
C. $\begin{bmatrix} -1 \\ -3 \end{bmatrix}$
D. $\begin{bmatrix} 1 \\ -3 \end{bmatrix}$

16. If $AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, then $B^t A^t$ is equal to

A. $\begin{bmatrix} -1 & 3 \\ 2 & -4 \end{bmatrix}$

B. $\begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$

C. $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

D. $\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$

17. If $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then $|Adj\ A|$ is equal to

A. -1
B. 0
C. 1
D. 2

18. By using Cramer's rule the value of y for the system of equations $x - y = 3$ and $2x + 5y = 4$ is equal to

A. $\dfrac{\begin{vmatrix} 1 & -3 \\ 2 & -4 \end{vmatrix}}{\begin{vmatrix} 1 & -1 \\ 2 & 5 \end{vmatrix}}$

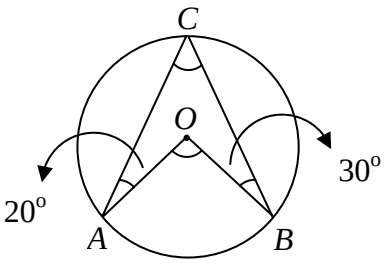
B. $\dfrac{\begin{vmatrix} 1 & -1 \\ 2 & 5 \end{vmatrix}}{\begin{vmatrix} 1 & -1 \\ 2 & 5 \end{vmatrix}}$

C. $\dfrac{\begin{vmatrix} 3 & -1 \\ 4 & -5 \end{vmatrix}}{\begin{vmatrix} 1 & -1 \\ 2 & 5 \end{vmatrix}}$

D. $\dfrac{\begin{vmatrix} 1 & 3 \\ 2 & 4 \end{vmatrix}}{\begin{vmatrix} 1 & -1 \\ 2 & 5 \end{vmatrix}}$

19. In the given figure O is the centre of the circle. If $\angle CAO = 20°$ and $\angle CBO = 30°$, then $\angle AOB$ is equal to

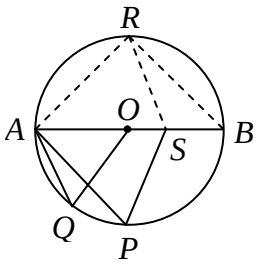
A. 25°
B. 80°
C. 100°
D. 120°



NOT TO SCALE

20. If O is the centre of the circle as shown in the diagram, then which of the following angles represents a right angle?

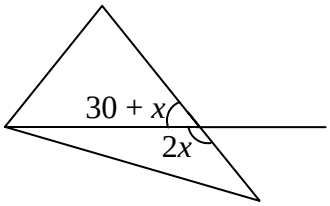
A. $\angle ARS$
B. $\angle AQO$
C. $\angle APS$
D. $\angle ARB$



NOT TO SCALE

21. The value of $2x$ in the given figure is equal to

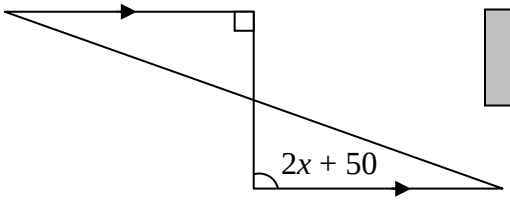
A. 30°
B. 50°
C. 100°
D. 120°



NOT TO SCALE

22. For the given figure the value of x is equal to

A. 20°
B. 40°
C. 70°
D. 140°



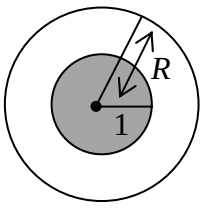
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23. In which of the following types of triangles all the three altitudes are concurrent at the vertex?

A. An acute angled triangle
B. An obtuse angled triangle
C. A right angled triangle
D. An equilateral triangle

24. For the given figure the area of the un-shaded region is equal to

A. πR^2
B. $\pi\left(R^2 - 1\right)$
C. $\pi\left(R+1\right)^2$
D. $\pi\left(R-1\right)^2$



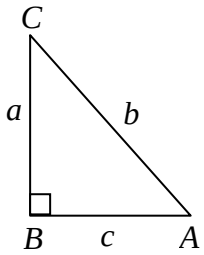
NOT TO SCALE

25. If the length of the side of a cube is 4 cm, then the volume in cm^3 is equal to

A. 12
B. 16
C. 24
D. 64

26. In the given right angled triangle if $a = c = x$, then the value of *b* is equal to

A. $\sqrt{2}\,x$
B. $2x$
C. x^2
D. x^4



NOT TO SCALE

27. If *x*, *y* and *z* are the measures of the sides of a right angled triangle , then which of the following also represents the measures of the sides of a right angled triangle?

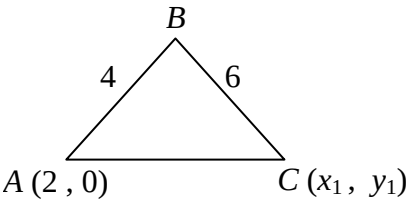
A. $-x, -y, -z$
B. $kx,\ ky, kz$
C. $x,\ y, \sqrt{2}z$
D. $3x, 4y, 5z$

28. If $A(a\ ,\ 3)$, $B(0\ ,\ 1)$ and $m\overline{AB}$ is equal to $\sqrt{4}$ units, then the value of *a* is equal to

A. – 3
B. – 1
C. 0
D. 1

29. In the given figure if *ABC* is an isosceles triangle, then the point $C(x_1, y_1)$ is

A. (4 , 0)
B. (6 , 0)
C. (0 , 6)
D. (0 , 4)



NOT TO SCALE

30. On Cartesian plane with reference to origin the point (3 , – 4) represents

A. 3 units left and 4 units down.
B. 3 units right and 4 units down.
C. 3 units up and 4 units down.
D. 3 units up and 4 units left.

END OF PAPER

Please use this page for rough work